

Platform Architecture Overview — v5

Working title: School Intelligence Platform **For:** the internal team and development partners · Draft 5.0 **What changed from v4:** the model is re-expressed on two axes (capabilities × domains) so "engine" and "area" stop colliding; the engines are re-tiered (3 domain engines + 2 platform services + the intelligence layer); the **semantic / metric layer** is named as a first-class component because it is the part that makes "intelligence" real; and a **market-positioning** section is added, grounded in a live competitive scan (June 2026).

1. The model in one idea (unchanged — it's right)

Every school already runs accounting, an SIS, payroll, HR, donor management, document storage, spreadsheets, email and calendars. The problem is not a shortage of software; it is that **none of these systems work together to help leaders run the institution.**

The platform does not replace those systems. It **orchestrates** them: it sits above the tools a school already uses, connects their information, captures decisions, moves work along, keeps the institution's knowledge in one place, and gives leaders a single place to run the school from. The clearest expression of that is the **digital COO** — the screen a leader sees first thing Monday morning that has already done the prioritising: *"Three things require your attention today."*

Everything else in this document exists to make that briefing **accurate, current, and worth acting on.**

2. The structural fix: two axes, not one list

v4 carried two overlapping vocabularies — **5 engines** and **8 areas the platform knows about** — and they didn't line up: finance, governance and planning appeared in both, while enrollment, HR, facilities and advancement were "areas" with no engine. That ambiguity is resolved by separating the two things they actually are:

- **Capabilities — the engines ("how").** What kind of work the platform does: *connect & normalise · plan · govern · route work · remember · sense & decide.*
- **Domains — the eight areas ("what").** The subjects that work is done *to*: **finance, enrollment, HR, facilities, governance, advancement, accreditation, strategic planning.**

Every feature is a cell in a capability × domain grid. "Budget vs. actual" = *Plan × finance*. "Policy review cycle" = *Govern × governance*. "Accreditation evidence captured

from daily operations" = *Remember* × *accreditation*. This single reframe removes almost all of the naming confusion in v4 and gives the team a precise way to place any new idea.

Capability ↓ \ Domain →	Finance	Enrollment	HR	Facilities	Governance	Advancement	Accreditation	Stra
Connect & normalise	✔ wedge	□	□	□	—	□	—	—
Plan	✔ budget	□	□	□	—	□	—	□
Govern	—	—	—	—	□	—	□	□
Route work (Workflow)	runs across every domain →							
Remember (Knowledge)	runs across every domain →							
Sense & decide (Intelligence)	reads across every domain → produces the briefing							

(✔ built today · □ planned · — system-of-record lives inside the platform / N/A. See §8 for the real build state.)

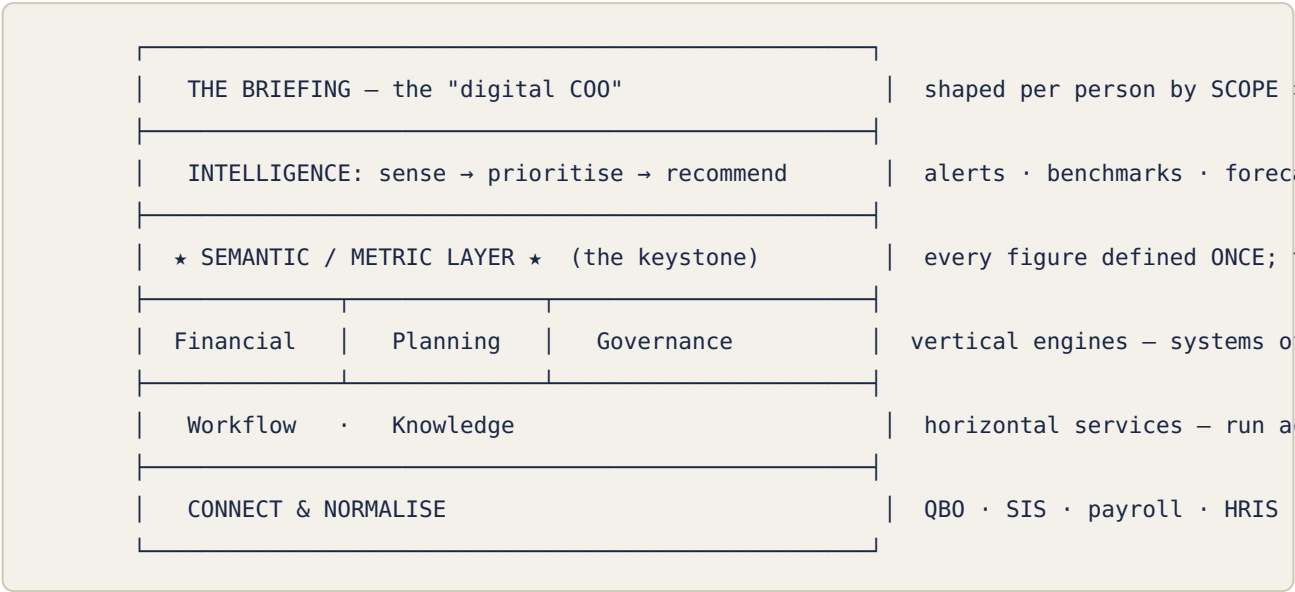
3. The engines, re-tiered

The five v4 engines are **not peers**. Three own a domain and are the system of record; two are horizontal services every domain uses; and intelligence is the layer on top, not a sibling.

Tier	Engine	Role	System of record?
Vertical (domain) engines	Financial · Planning · Governance	own a domain end-to-end	yes — these live inside the platform
Horizontal (platform) services	Workflow · Knowledge	run across <i>all</i> domains	n/a — they serve the others
Top layer	Intelligence	senses across everything and decides	n/a — it reads, it doesn't store

For the four domains the platform does **not** own — enrollment, HR, facilities, advancement — it **reads from and writes back to** the systems the school already runs (SIS, HRIS, facilities, donor CRM). Either way the information lands in one place. (This "system of record vs. connector" split was implicit in v4; making it explicit is what lets us say honestly which domains we build vs. integrate.)

4. The layer cake (the structure, top to bottom)



Modules vs. engines. What a user clicks — *Finance, Budget, Statements, Reports, Governance, the morning briefing* — are **modules**: the navigable app surface rendered *from* the engines. Engines are the platform beneath. (This is the "intelligence section with modules under it" instinct, made precise: the briefing is the top module; each engine projects one or more modules.)

5. The semantic layer — the keystone, and the actual moat

The starred row is the part v4 was missing, and it is the hardest and most valuable piece.

The platform's core promise is that **two people never see figures that disagree**. That only holds if there is **one canonical place that defines each figure once** — "cash position," "enrollment vs. plan," "budget variance," "days cash on hand" — so every engine, every module and every personalised briefing reads the *same* number from the *same* definition. Concretely the semantic layer is three things:

1. **A canonical data model** — normalised entities (school, period, account, fund, person, plan, decision) that all sources map into.
2. **A metric/indicator registry** — each KPI defined once, with its formula, inputs, thresholds and owning domain, versioned and traceable back to source (the audit trail of *meaning*, paralleling the financial audit trail).
3. **A prioritisation/alert engine** — rules and models that turn metrics into *ranked attention* ("below target in November," "behind schedule") — which is what the Intelligence layer reads to assemble the briefing.

Why this is the moat, not any single engine: every individual engine has a credible competitor (see §7). What no point-solution can replicate is a *connected* metric layer spanning domains, because they only ever see one domain. Build the engines for coverage; build the semantic layer for defensibility.

6. The briefing changes with the reader (Scope × Lens)

The Monday-morning screen is not one view for everyone. From the same connected figures, the Intelligence layer assembles a per-person briefing along two dimensions:

- **Scope** — how much of the organisation you're responsible for (one school → an organization's finances → the whole system). Scope is also a **boundary**: you see only what your role entitles you to, following the org hierarchy the solution pack defines.
- **Lens** — which decisions and which of the eight domains sit with you (a finance director's briefing centres on money; a principal's on running their school; a superintendent's reaches across all eight).

Same engines, same figures, three different briefings — Principal · Organization finance director · Superintendent. This is the feature competitors can't fake without the semantic layer underneath it.

7. Packaging — a core platform + licensable modules

The platform must be **modular**: a school that only wants finance can buy *just* finance, with **price tiers per module**, and each module must run on its own *or* plug into the whole and add capability. The good news is the engine decomposition was already built for this ("each engine independently valuable" is a founding rule). Per-module licensing turns that implicit modularity into an explicit split:

The core (always shipped — the substrate a module plugs into; cannot be unbundled): - identity / RBAC / org hierarchy - the **semantic / metric layer** (§5) - the **intelligence / briefing shell** (§4, §6) - the horizontal services — **Workflow** and **Knowledge**

The modules (independently licensed; each runs standalone with just core + itself): - the vertical engines — **Finance, Planning, Governance** - the read-domain connectors — **Enrollment, HR, Facilities, Advancement, Accreditation**

A school can license **only Finance** and get a working product (core + Finance, with a finance-only briefing). Add modules and the core *composes* them.

The plug-in contract (how a module attaches)

Each module is pluggable because it **registers what it contributes** to the core, rather than the core hard-coding it:

A module declares...	...which the core composes into
metrics (formula, thresholds, scope-aggregation)	the semantic/metric registry (§5)
alerts / briefing items	the prioritisation engine → the briefing (§6)
task & approval types	Workflow
document / evidence types	Knowledge
a nav surface + routes	the app shell
an entitlement key	billing / feature-gating (below)

Enabling or disabling a module (by license tier) lights up or hides exactly its metrics, briefing items, nav and routes — nothing else needs to know.

Per-module entitlement (extend the billing you already have)

Today entitlement is **binary**: `isEntitled(schoolId) → true/false`, one `plan` per school, `EntitlementGuard → 402`. Per-module pricing is an *extension*, not a new system: - `Subscription` carries a **set of licensed modules** (+ a tier per module), not one plan. - `isEntitled(school, module)` and a **module-aware guard** (each module's routes declare the entitlement key they need). - per-module **Stripe prices**; the core is the base.

The commercial payoff: modularity and the briefing reinforce each other

A finance-only customer gets a **finance-only COO briefing** — valuable alone. Add Enrollment → the briefing becomes cross-domain ("enrollment down *and* cash below target"). Add Governance → fuller COO. **The intelligence layer's value scales with how many modules you own — that's the built-in upsell / land-and-expand engine.** Per-module pricing isn't fighting the architecture; it's powered by it.

How to build it (don't over-engineer)

Do a **modular monolith**, not a runtime plugin system: clean module boundaries (largely already there — one NestJS module + engine per module), the registration contract above, and **entitlement-gated feature flags** per module. That buys independent licensing + standalone operation + plug-in composition *without* microservices/plugin-runtime cost — right for a small team. The contract makes extracting a module into its own service *later* cheap, if scale ever demands it.

8. Where we sit in the market (live scan, June 2026)

A multi-source competitive sweep (sources cited in the companion research file) supports a clear conclusion: **the integrated cross-domain "digital COO" briefing for schools is genuinely unclaimed — but the gap is narrowing, and the threat is not who we assumed.**

Layer	Who's there	What they do	The gap they leave
Suite incumbent	Blackbaud (Financial Edge NXT, Raiser's Edge NXT, SIS, Billing, BoardEffect)	discrete products + <i>point-to-point</i> integrations; per-product "Copilot" AI	sold as separate products; tightest integration is only "intelligent reporting," no unified cross-domain briefing ; consolidation lives on separate pages
SIS + billing (and the real threat)	FACTS / Nelnet — building FACTS IQ	branded "the intelligence layer," "central nervous system," "single system of record," "30+ decision-ready dashboards"	single-school, SIS-centric; only 4 hubs (admissions/engagement/financial/success); no HR, facilities, governance, accreditation, strategy, or multi-school consolidation; dashboards, not a prioritised briefing
	Veracross	connected single-school SIS+billing+accounting	accounting positioned as an <i>efficiency</i> tool — no leadership/briefing layer
Finance / FP&A	Frontline/ Forecast5, Sage Intacct, Adaptive/ Vena/Prophix	strong multi-year forecasting & multi-entity GL	finance-domain only ; Forecast5 is public-district only
Multi-school consolidation	ParishSOFT (parish-focused), separate FE NXT pages	parish/diocese financial reporting	multi-school (multi-entity) consolidation is largely unclaimed by the SIS/suite incumbents on their core pages
Governance	BoardEffect, OnBoard, Diligent, Boardable, BoardPro	board portals	governance domain only
AI-era entrants ('25-'26)	schoolOS (IT/ops "operational intelligence"), Scout (YC W25, AI SIS)	orchestrate-not-replace, but for IT/ops and SIS busywork	adjacent slices, not the cross-domain COO briefing

Verdict: - **(a) White space is real (high confidence).** No verified player ships a prioritised, role-shaped, cross-domain daily *briefing* across all eight domains. - **(b) Single biggest threat: FACTS (Nelnet).** It already sits on enrollment + tuition + finance data

and is actively shipping an "intelligence layer." But its scope is bounded to its own SIS ecosystem, single-school, and it markets **dashboards, not prioritisation** — and it has **no governance/accreditation/strategy and no multi-school consolidation**. Our three durable differentiators are exactly its three gaps: **(1) a prioritised briefing, not dashboards; (2) full eight-domain orchestration; (3) multi-school (multi-entity) consolidation across districts, networks, systems and dioceses.** - **(c) Near-analogue worth studying: schoolOS** (same "orchestrate, don't replace" thesis, but for IT/operations) and higher-ed **student-success/early-alert** tools (EAB Navigate, Civitas) that nail the "prioritised morning briefing" pattern — for *students*, not institutional operations. Borrow the pattern; own the domain they don't.

Strategic read: lead go-to-market with the two unclaimed things — **the briefing** and **multi-school consolidation** — using **finance as the proof**. The core market is **multi-school organizations** — districts, networks, systems and dioceses — where finance + multi-school consolidation pain is sharpest. FACTS will likely add domains and prioritisation over time; the race is to own the cross-domain *leadership* layer (and the multi-school consolidation wedge) before they extend out of the SIS.

9. Build state & phased roadmap (mapped to what exists)

Honest inventory of the current codebase against the model:

Engine / layer	Built today	Gap
Financial (vertical)	~80% — Data hub (sources incl. QBO), mapping/normalisation, Statements (validation/reconciliation + reporting), Budget (+ forecast, budget-vs-actual, multi-school roll-up), Monthly actuals, Reports + Board packets , Readiness/Compliance, period history	deepen QBO sync; harden multi-school consolidation as a headline
Intelligence + briefing	seeded — Analytics (dashboards/metrics/benchmarks), Penny (AI assistant + autonomous actions), Home (briefing surface)	it only reads the <i>finance</i> domain; no canonical metric registry; no prioritisation → not yet a real cross-domain briefing
Semantic / metric layer	implicit — each module computes its own numbers	not unified — this is the keystone work
Planning (vertical)	partial — Budget + forecast (the finance slice)	strategic plan, improvement plans, facilities/capital, progress tracking
Governance (vertical)	board packets only	boards/committees, meetings/agendas, minutes/decisions, policy register
Workflow (horizontal)	thin — Schedules, report-schedule (recurring)	tasks, approvals/sign-offs, reminders, document routing
Knowledge (horizontal)	thin — Readiness (accreditation-evidence seed)	document store, templates, institutional memory, search

The roadmap (depth on the differentiator before breadth)

Phase 0 — Finish the wedge (*in progress*). Financial intelligence engine + multi-school roll-up as a *headline*, deeper QBO sync. This already delivers value alone and earns the trust + data everything else depends on. ✅ largely done.

Phase 1 — Build the semantic layer and the *real briefing* (*do this next — it's the differentiator*). Extract the metric/indicator registry + prioritisation engine from the finance domain; turn Home + Penny into an actual *prioritised* briefing ("3 things today"), finance-only at first. **This is the demo that wins, and it must come before more engines** — it's what converts "another finance tool" into "the digital COO," and it's precisely what FACTS's dashboards are not.

Phase 2 — First non-finance domain into the briefing. Add **enrollment** (read from the SIS) — because "*enrollment down 6 vs. plan*" + "*cash below target in November*" in one briefing is the cross-domain moment no competitor delivers, and it's directly on FACTS's

turf where they still lack prioritisation. (Enrollment is also the lowest-integration-cost first expansion.)

Phase 3 — Governance + Workflow primitives. Grow **Governance** out from the existing board-packets into meetings/minutes/policies; this *requires* the first real **Workflow** primitives (tasks, approvals, reminders, document routing). Multi-school organizations and boards feel this acutely, and it deepens the governance white space competitors leave open.

Phase 4 — Knowledge + the read-only domains. Stand up the **Knowledge** engine (document store, accreditation evidence promoted from Readiness, search, institutional memory) and connect the remaining read-from-external domains — **HR, facilities, advancement** — as feeds into the briefing.

Cross-cutting — solution packs. The platform's identity is general; each market gets a pack — **independent schools, charter schools, private-school networks, dioceses, small colleges** — as configuration (org hierarchy, compliance rules, templates, report packages) over the same architecture. No single pack leads; the common wedge across all of them is **finance + multi-school consolidation**, the unclaimed white space.

Sequencing principle: hold every engine to "independently valuable before the next starts" (finance already passes). Resist the everything-platform sprawl — the failure mode is *shallow everywhere*. Win by being **deep on finance + the briefing**, then expand one domain at a time, each one making the briefing smarter.

10. The one test (unchanged)

Every feature, on any engine, must answer one question: **does this help leaders make better decisions?** If a proposed addition doesn't save time, improve a decision, strengthen accountability, preserve institutional knowledge, or reduce risk, it doesn't belong in the platform. Add one corollary from §8: *and is the engine it sits on already independently valuable, or does this feature make it so?*